

PanoAI–Dentist: A multi–expert artificial intelligence framework for automated dental lesion diagnosis and report generation from panoramic radiographs

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Abstract

Purpose: To develop PanoAI–Dentist, a multi–expert artificial intelligence system designed to automate dental diagnosis and structured clinical report generation from panoramic radiographic images.

Materials and Methods: The proposed system integrates four task–specific vision experts for periodontitis staging, jaw tumor and cyst segmentation, caries detection, and periapical lesion localization. A large language model–based Report Writer expert consumes structured JSON outputs from these vision experts and generates clinically formatted XML reports following Seoul National University Dental Hospital reporting guidelines. Vision experts were implemented using SOTA architectures including AttUNet, UNet++, SegFormer, RT–DETR–l, YOLOv8m, and YOLO26m, with the best–performing model selected for each task.

Results: Quantitative evaluation demonstrated strong and consistent performance across all tasks. SegFormer achieved the highest Dice scores for periodontitis staging (0.898) and caries segmentation (0.624), while AttUNet performed best for tumor/cyst segmentation (0.816). YOLOv8m yielded the best detection performance for periapical lesions (mAP@0.5: 0.694). Report generation achieved high agreement with reference reports, with BLEU, METEOR, ROUGE–L, and BERTScore (F1) scores of 92.12, 0.9674, 0.9044, and 0.9867, respectively.

Conclusion: PanoAI–Dentist demonstrates the feasibility of integrating multi–expert vision models with a LLM–based report generation system to produce automated, structured dental diagnostic reports from panoramic radiographs with high diagnostic and reporting accuracy. (*J Artif Intell Dent* 2026; 2(1): 13–20)

KEY WORDS: Dentistry; Deep Learning; Large Language Model; Diagnostic Imaging; Image Interpretation, Computer Assisted

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Introduction

Dental diseases such as periodontitis, caries, periapical lesions, and jaw tumors affect a large proportion of the global population and represent a significant burden on oral healthcare systems. Panoramic radiography is a widely used, cost–effective imaging modality in clinical dentistry that provides a comprehensive view of the dentition, alveolar bone, and surrounding anatomical structures. However, accurate interpretation of panoramic radiographs demands substantial expertise, and inter–clinician variability in diagnosis remains a persistent challenge.

Recent advances in deep learning have demonstrated remarkable potential for automating diagnostic tasks in

medical imaging. Convolutional neural networks and transformer-based architectures have been successfully applied to detect, segment, and classify pathological findings in dental radiographs. Nevertheless, most prior work focuses on individual conditions in isolation, and few systems integrate multi-disease diagnostic capabilities into a unified clinical workflow.

In this paper, we introduce PanoAI-Dentist, a multi-expert AI framework that orchestrates four specialized vision experts and a large language model (LLM)-based Report Writer expert. Given a panoramic radiograph, each vision expert independently performs disease-specific analysis and returns structured JSON findings. The Report Writer expert synthesizes these findings into a structured XML clinical report consistent with institutional formatting guidelines.

Our contributions are threefold: (1) we develop and benchmark task-specific vision experts for periodontitis staging, tumor and cyst segmentation, caries detection, and periapical lesion detection using state-of-the-art architectures; (2) we design an LLM-based report generation pipeline capable of producing human-readable clinical narratives from structured diagnostic JSON; and (3) we provide evaluation of all the four vision experts and the report generation system across four disease categories.

Materials and Methods

1. System overview

PanoAI-Dentist consists of two principal components: 1) a set of four task-specific vision experts, and 2) a Dental Report Writer expert. Upon receipt of a panoramic radiograph, the system activates the appropriate expert module for each disease domain and routes the structured outputs to the report generation pipeline (Fig. 1).

2. Vision experts

The system includes four task-specific vision experts, each denoted for disease category. For periodontitis, periodontitis expert (E_{PD}) segments teeth, cemento-enamel junction (CEJ), and alveolar bone levels, then computes tooth-wise stage labels by measuring the vertical difference between junction and bone levels on a four-level scale from normal to severe. For tumor and cyst analysis, the expert ($E_{T/C}$) performs multi-class segmentation producing pixel-wise masks for ameloblastoma, periapical cyst, dentigerous cyst, and odontogenic keratocyst. For caries detection, a pre-trained tooth instance segmentation model (M_S) is first used to delineate individual teeth (Dice: 0.96 ± 0.18 , intersection over union: 0.83 ± 0.17), after which the caries expert (E_C) analyzes tooth-level crops to localize carious regions. Periapical lesions are first detected using object detection to generate bounding boxes for candidate regions by the periapical expert (E_{PA}). Then, the overlap between each bounding box and the segmented tooth instance masks is computed to assign the corresponding tooth number to each detected lesion.

Segmentation experts were implemented using AttUNet [1], UNet++ [2], and SegFormer [3]. Detection experts were evaluated using RT-DETR-l [4], YOLOv8m [6], and YOLO26m [7]. The best-performing architecture was selected based on validation performance per task. All experts output structured JSON representations consumed by the Report Writer expert.

3. Report writer expert

The Report Writer expert is powered by Qwen3-32B [5], loaded in float16 precision on an NVIDIA RTX PRO 6000 Blackwell GPU (NVIDIA, Santa Clara, CA, USA) with 96 GB memory. The expert consumes structured JSON diagnostic findings from activated vision experts and produces a structured XML clinical report with two primary

sections: a [reading] section containing radiographic observations, and a [conclusion] section containing diagnostic interpretation.

Report generation strictly follows the clinical reporting and formatting guidelines of Seoul National University Dental Hospital. The guidelines are provided to the LLM as instructional prompts, with task-specific prompts assigned to each expert output to guide the generation of the corresponding diagnostic report.

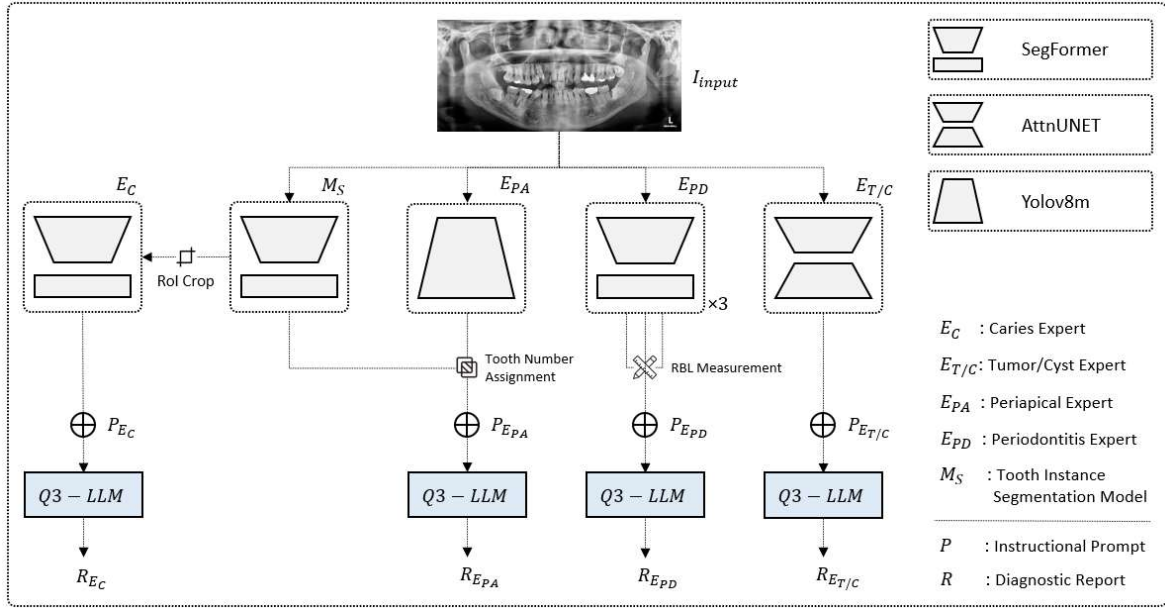


Fig. 1. Overview of the proposed multi-expert diagnostic framework. A panoramic radiograph is processed by four task-specific vision experts for caries, tumor/cyst, periapical lesions, and periodontitis. The expert outputs are combined with instructional prompts and provided to the Qwen3 (Q3-LLM) for diagnostic report generation.

4. Datasets and training

The periodontitis dataset includes 75 images with tooth, CEJ, and bone level annotations for staging. The jaw tumor and cyst dataset include 957 images across four categories confirmed by histopathological and radiological consensus. The periapical lesion dataset includes 571 images annotated with bounding boxes. The public caries dataset comprises 597 radiographs yielding 5,780 tooth-level crops [8]. Except for caries dataset (DC1000), all datasets are in-house and were annotated by specialists and double-validated to ensure labeling quality. All datasets were randomly divided into training and test sets with an 8:2 ratio.

Since no model training was performed and the report generation relied solely on instructional prompting, the evaluation was conducted on a test set of 40 cases (10 cases per lesion category). This design was constrained by the limited availability of periodontitis data; therefore, to avoid class imbalance in the test set, an equal number of cases was selected for each lesion category.

All experts were implemented in PyTorch and trained on an NVIDIA RTX A6000 GPU with 48 GB memory. Models were trained for up to 500 epochs with a batch size of 8 using Adam optimizer with an initial learning rate of 1×10^{-4} . A ReduceLROnPlateau scheduler with patience 10 was applied, and early stopping was triggered after 20 epochs without improvement.

Results

1. Vision expert performance

Table 1 presents quantitative performance of all evaluated architectures across dental diagnostic tasks. For periodontitis staging, SegFormer achieved the highest Dice score (0.898 ± 0.057) and precision (0.917 ± 0.063) [9]. Although evaluated on a limited dataset, staging was further supported by RBL-based post-processing, classifying stages 1–3 based on predefined thresholds ($<15\%$, $15\text{--}33\%$, $>33\%$), while stage 4 was excluded due to lack of clinical data beyond panoramic radiographs.

Table 1. Quantitative performance of architectures across dental diagnostic tasks

	Model	Dice	IoU	Precision	Recall
Periodontitis staging	UNet++	0.892 ± 0.068	0.832 ± 0.082	0.897 ± 0.070	0.908 ± 0.057
	AttUNet	0.894 ± 0.079	0.835 ± 0.092	0.905 ± 0.078	0.901 ± 0.066
	SegFormer*	0.898 ± 0.057	0.832 ± 0.070	0.917 ± 0.063	0.892 ± 0.052
Tumor/cyst segmentation	SegFormer	0.791 ± 0.166	0.776 ± 0.153	0.922 ± 0.095	0.851 ± 0.111
	UNet++	0.800 ± 0.167	0.782 ± 0.154	0.915 ± 0.095	0.857 ± 0.109
	AttUNet*	0.816 ± 0.149	0.798 ± 0.136	0.932 ± 0.085	0.859 ± 0.111
Caries segmentation	UNet++	0.598 ± 0.451	0.570 ± 0.449	0.655 ± 0.437	0.866 ± 0.296
	AttUNet	0.616 ± 0.448	0.588 ± 0.446	0.680 ± 0.429	0.867 ± 0.294
	SegFormer*	0.624 ± 0.444	0.596 ± 0.444	0.690 ± 0.423	0.862 ± 0.299
	Model	mAP50	mAP50–95	Precision	Recall
Periapical lesion detection	RT–DETR–l	0.563	0.253	0.629	0.606
	YOLO26m	0.615	0.309	0.709	0.555
	YOLOv8m*	0.694	0.334	0.849	0.512

IoU: intersection over union, mAP: mean average precision. * Denotes selected model per task. Periodontitis results are averaged across teeth, CEJ, and bone level segmentations

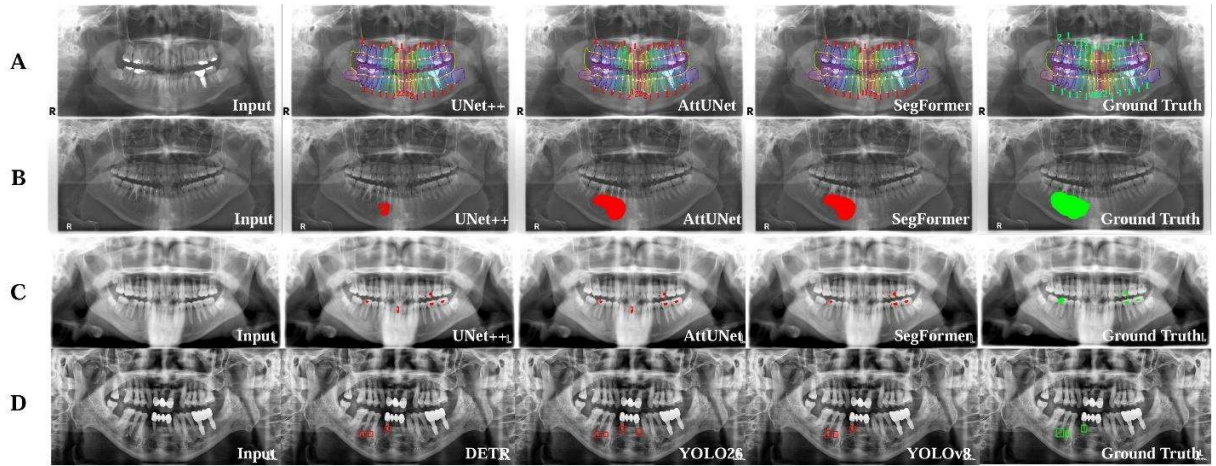


Fig. 2. Representative outputs of four vision experts. A. Periodontitis staging. B. Cyst and tumor segmentation. C. Caries segmentation. D. Periapical lesion detection.

For tumor and cyst segmentation, AttUNet outperformed both SegFormer and UNet++ with a Dice score of 0.816 ± 0.149 and IoU [10] of 0.798 ± 0.136 , demonstrating superior lesion delineation. For caries segmentation, SegFormer achieved the highest Dice (0.624 ± 0.444) and IoU (0.596 ± 0.444). For periapical lesion detection, YOLOv8m surpassed both RT-DETR-L and YOLO26m with mAP@0.5 of 0.694 and precision [11] of 0.849 (Fig. 2).

2. Report generation performance

Table 2 summarizes report generation quality across all four disease categories. The Tumor/Cyst category achieved perfect scores across all metrics (BLEU [12]: 100.00, METEOR [13]: 1.0, ROUGE-L [14]: 1.0, BERTScore [15] F1: 1.0), indicating exact reproduction of expected clinical narrative structure. Periapical lesion reports also attained near-perfect performance (BLEU: 98.96, BERTScore F1: 0.9994).

Table 2. Report generation performance across disease categories. R-L, BS-P and BS-F1 denote ROUGE-L, BERTScore Precision and F1, respectively

Category	BLEU	METEOR	R-L	BS-P	BS-F1
Caries	88.95	0.9519	0.8364	0.9831	0.9684
Periapical	98.96	0.9954	0.9960	0.9992	0.9994
Periodontitis	80.57	0.9223	0.7852	0.9792	0.9790
Tumor/Cyst	100.00	1.0000	1.0000	1.0000	1.0000
Overall	92.12	0.9674	0.9044	0.9904	0.9867

Overall represents the mean across all four categories (data number = 40).

Periodontitis yielded the most challenging report generation task with BLEU of 80.57 and ROUGE-L of 0.785, reflecting the greater linguistic complexity of staging-based reports. Caries reports achieved intermediate performance (BLEU: 88.95, ROUGE-L: 0.836). The overall system achieved a BLEU score of 92.12 and BERTScore F1 of 0.9867, demonstrating high semantic fidelity between AI-generated and reference clinical reports (Fig. 3).

$R_{ET/c}: GT$	$R_{ET/c}: Prediction$
<reading> * osseous lesion: jaw region </reading> <conclusion> r/o) periapical cyst </conclusion>	<reading> * osseous lesion: jaw region </reading> <conclusion> r/o) periapical cyst </conclusion>

Fig. 3. Qualitative comparison of the diagnostic reports generated by the Qwen3-32B model for the Tumor/Cyst expert outputs.

Evaluation was conducted on 40 cases (10 per lesion category), with balanced sampling due to limited

periodontitis data. To assess LLM performance independently of input noise, 10 healthy cases with high-confidence annotations were additionally selected. This was necessary as, in the prompt-based setting, report quality is highly dependent on the correctness of the input annotations.

Discussion

This study presents PanoAI-Dentist, a multi-expert AI framework that integrates four task-specific vision experts with an LLM-based report generation pipeline for automated dental diagnosis and structured clinical report generation from panoramic radiographs. The results demonstrate strong and consistent performance across all disease categories evaluated, and the report generation module produces clinically structured narratives of near-expert quality, supporting the feasibility of deploying such systems in real-world dental practice.

Among the vision experts, SegFormer demonstrated superior segmentation performance for both periodontitis staging (Dice: 0.898 ± 0.057) and caries segmentation (Dice: 0.624 ± 0.444), while AttUNet achieved the best results for tumor and cyst segmentation (Dice: 0.816 ± 0.149). For periapical lesion detection, YOLOv8m outperformed both RT-DETR-L and YOLO26m with an mAP@0.5 of 0.694, confirming the suitability of task-specific architecture selection in multi-disease diagnostic systems. The report generation pipeline, powered by Qwen3-32B, achieved an overall BERTScore F1 of 0.9867, with perfect scores for the tumor/cyst category and near-perfect performance for periapical lesions, demonstrating high semantic fidelity between AI-generated and reference clinical reports.

Periodontitis staging yielded the most challenging report generation results, with BLEU of 80.57 and ROUGE-L of 0.785, reflecting the inherent linguistic complexity of staging-based clinical narratives that require description of tooth-wise bone loss levels across multiple teeth. Caries report generation achieved intermediate performance (BLEU: 88.95, ROUGE-L: 0.836), consistent with the moderate segmentation accuracy of the caries expert. These findings indicate that report generation quality is closely coupled with the precision and richness of the structured JSON inputs produced by the vision experts, particularly in the prompt-based framework adopted in this study.

Several limitations should be acknowledged. The evaluation of the report generation module was conducted on a relatively small but balanced test set of 40 cases, constrained primarily by the limited availability of periodontitis data. To mitigate class imbalance, an equal number of cases was selected per lesion category; however, this design may not fully reflect the distribution of disease severity encountered in clinical settings. Additionally, periodontitis staging was restricted to stages 1–3 based on radiographic bone loss, as stage 4 requires additional clinical information—such as tooth mobility and masticatory dysfunction—that cannot be derived from panoramic radiographs alone. These constraints highlight the importance of acquiring larger, more diverse annotated datasets for future model development and evaluation.

Future work will focus on improving the relatively modest performance in caries segmentation (Dice: 0.624), for example by leveraging recent advances in deep learning-based caries segmentation on panoramic radiographs that report higher accuracy and more robust multistage lesion delineation [16,17].

Future work will also include prospective clinical validation studies in collaboration with practicing dentists, building on prior work that has evaluated AI systems alongside clinicians in real-world diagnostic workflows [18,19].

In addition, we plan to extend the system to additional imaging modalities (e.g., periapical radiographs and CBCT) and disease or treatment categories, guided by recent multimodal dental datasets and AI platforms that support diverse diagnostic tasks in clinical environments [20,21].

Conflicts of Interest: None

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